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To cite this article: Yu Song, Ling Li, Jacob Layer, Raychl Fairbanks, Gerwyn Hughes, Derek Smith, Margaret Wilson, Qin Zhu & Boyi Dai (2024) Unanticipated mid-flight external trunk perturbation increased frontal plane ACL loading variables during sidestep cuttings, Journal of Sports Sciences, 42:7, 599-610, DOI: [10.1080/02640414.2024.2353404](https://doi.org/10.1080/02640414.2024.2353404)

To link to this article: <https://doi.org/10.1080/02640414.2024.2353404>



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Unanticipated mid-flight external trunk perturbation increased frontal plane ACL loading variables during sidestep cuttings

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ABSTRACT

Unanticipated trunk perturbation is commonly observed when anterior cruciate ligament (ACL) injuries occur during direction-changing manoeuvres. This study aimed to quantify the effect of mid-flight medial-lateral external trunk perturbation directions/locations on ACL loading variables during sidestep cuttings. Thirty-two recreational athletes performed sidestep cuttings under combinations of three perturbation directions (no-perturbation, ipsilateral-perturbation, and contralateral-perturbation relative to the cutting leg) and two perturbation locations (upper-trunk versus lower-trunk). The pushing perturbation was created by customised devices releasing a slam ball to contact participants near maximum jump height prior to cutting. Perturbation generally resulted in greater peak vertical ground reaction force and slower cutting velocity. Upper-trunk contralateral perturbation showed the greatest lateral trunk bending away from the travel direction, greatest peak knee flexion and abduction angles, and greatest peak internal knee adduction moments compared to other conditions. Such increased ACL loading variables were likely due to the increased lateral trunk bending and whole-body horizontal velocity away from the cutting direction caused by the contralateral perturbation act at the upper trunk. The findings may help understand the mechanisms of indirect contact ACL injuries and develop effective cutting techniques for ACL injury prevention.

ARTICLE HISTORY

Received 12 December 2023
Accepted 2 May 2024

KEYWORDS

ACL injuries; anterior cruciate ligament; kinematics; kinetics; biomechanics

Introduction

An injury to the anterior cruciate ligament (ACL) typically occurs shortly after initial ground contact (IC) in direction-changing manoeuvres (Hughes & Dai, 2023; Koga et al., 2010; Song, Li, Hughes, et al., 2023). ACL injury negatively impairs patients' quality of life in the long term and can create a considerable financial burden to society (Mather et al., 2013; Poulsen et al., 2019). Previous video analyses have demonstrated that the ACL commonly tears in a single-leg landing pattern, characterised as the injured leg experiencing a high proportion of the total ground reaction force (GRF) with the knee in a near-fully extended position when the player tends to pass an opponent or avoid a collision by changing directions sideways (Johnston et al., 2018; Koga et al., 2010; Olsen et al., 2004; Song, Li, Hughes, et al., 2023). Such knee positions observed in ACL injuries are consistent with ACL loading mechanisms identified by *in vivo* and *in vitro* studies (Beaulieu et al., 2023; Boden & Sheehan, 2022; Englander et al., 2022).

In addition to knee mechanics (limited knee flexion angle, increased knee abduction, and internal rotation angles, as well as greater internal knee extension, adduction, and external rotation moments), trunk motion and foot placement play an important role in ACL injuries during direction-changing

manoeuvres, such as cross and sidestep cutting tasks (Dempsey et al., 2007; Donelon et al., 2020; Kristianslund et al., 2014). Dempsey et al. (Dempsey et al., 2007). showed that lateral trunk bending away from the travel direction led to a greater peak internal knee adduction moment (the internal adduction moment of the knee joint that needs to be produced to counterbalance the external knee abduction moment imposed on the knee joint) in sidestep cutting, which is associated with increased ACL loading. Moreover, the greatest knee moments were reported in the technique of the foot laterally being placed away from the travel direction. Kristianslund et al. (Kristianslund et al., 2014) found that greater lateral trunk bending away from the travel direction, along with more laterally placed foot positions relative to the whole-body centre of mass (COM), were associated with increased internal knee adduction moment. Consistently, moving the COM medially and anteriorly towards the travel direction can effectively decrease internal knee adduction moment during sidestep cutting, which is associated with reduced ACL injury risk (Donnelly et al., 2012). While trunk motion aligned with foot placement appears to be associated with ACL loading variables in sidestep cuttings, previous studies only quantified self-initiated techniques without including unanticipated external

trunk perturbation. Given unanticipated trunk contact with opponents/objects in sports, such as being pushed or held prior to and/or near the estimated time of injury, is frequently observed in ACL injuries (Della Villa et al., 2020; Olsen et al., 2004; Song, Li, Hughes, et al., 2023), understanding the effect of external trunk perturbation on ACL loading variables during sidestep cutting is needed.

Previous laboratory studies determined the effect of trunk perturbation during jump-landings (Song, Li, Layer, et al., 2023, 2023; Yom et al., 2014). Mid-flight medial-lateral external pushing perturbation resulted in greater lateral trunk bending angles, knee abduction angles, and internal knee adduction moments for the leg contralateral to the perturbation direction during jump-landings, associated with increased ACL loading (Song, Li, Layer, et al., 2023, 2023). Additionally, the upper-trunk perturbation resulted in greater increases in ACL loading variables compared to the lower-trunk perturbation during double-leg landings (Song, Li, Layer, et al., 2023). Yet, limited studies assessed the effect of external trunk perturbation during cutting tasks. First, cutting is a common task associated with ACL injuries (Hughes & Dai, 2023). Second, cutting includes follow-up movements after IC, imposing additional performance demands, such as fast cutting velocities and short stance time. Such performance needs were correlated with increased ACL loading variables during sidestep cutting (Dai et al., 2019; Dos'Santos et al., 2021; Fox, 2018). For example, greater peak internal knee adduction moment is associated with greater peak knee abduction angles, greater peak vertical GRF (VGRF), greater peak internal knee external rotation moments, and shorter stance time during sidestep cuttings (Dos'Santos et al., 2021). Furthermore, faster cutting velocity resulted in shorter stance time, greater peak GRF, and greater peak internal knee extension moment (Dai et al., 2019; Dos'Santos et al., 2021). Therefore, identifying the biomechanical connections between external trunk perturbation and ACL loading variables, as well as cutting performance, will help explain the link between whole-body movement and the injured knee and provide knowledge in understanding indirect contact ACL injury mechanisms (Song et al., 2022).

This study aimed to quantify the effects of unanticipated mid-flight medial-lateral external perturbation direction (no/contralateral/ipsilateral perturbations) and location (upper-trunk/lower-trunk) on ACL loading variables and performance during sidestep cuttings. It was hypothesised that the contralateral-perturbation direction and the upper-trunk location would show greater ACL loading variables compared to ipsilateral-perturbation and no-perturbation directions and the lower-trunk perturbation location, respectively. Furthermore, contralateral perturbation to the upper trunk would result in decreased cutting velocities and increased stance time compared to other perturbation directions and locations.

Methods

Participants

The smallest effect sizes were 0.83 and 1.11 between perturbation and no-perturbation in peak VGRF during single-leg and double-leg landings, respectively, in previous studies (Song, Li,

Layer, et al., 2023, 2023). A sample size of 14 was needed to achieve a power of 80% at a Type I error rate of 0.05. Thirty-two recreational athletes (sex: 16 males and 16 females; age: 22.7 ± 3.1 years; height: 1.7 ± 0.1 m; mass: 71.3 ± 11.9 kg) participated in this study. Participants were recruited if they had experience playing sports that involved jump-landing activities (soccer, basketball, American football, etc.) and were physically active and/or exercising at least twice per week for a total of 2 hours at the time of testing. The exclusion criteria were having a previous ACL injury or any other lower limb surgery; having a lower limb injury that prevented participation in physical activities for more than 2 weeks in the last half year; having any conditions that prevent them from maximal effort in sports; being allergic to adhesive; or being pregnant or thinking they might be pregnant (Song, Li, Layer, et al., 2023). This study was approved by the University of Wyoming Institutional Review Board. Participants signed a consent form prior to participation.

Perturbation & task

The external perturbation was created by a pair of customised apparatuses releasing a 4.54 kg slam ball on each side (Song, Li, Layer, et al., 2023). The ball was designed to contact the participants' upper trunk (between the acromion and the 5th rib) or lower trunk (between the iliac crest and the greater trochanter) region near the maximal jump height with an approximate horizontal velocity of 5 m/s (Song, Li, Layer, et al., 2023). Participants were protected by helmets and two sponge pads placed over the thoracic cage during the testing. The task included in this study was a sidestep cutting. Participants stood with their feet shoulder-width apart and each foot on a force platform (FP4060–10, Bertec, 1600 Hz) and jumped vertically for a maximal height with both arms raised up during mid-flight. Participants landed with one pre-determined foot (counterbalanced order among participants) on a force platform and performed a 45-degree sidestep cutting as fast as possible for a minimum of 5 metres (Figure 1). A tape was placed on the ground to indicate the cutting direction. This task was designed to simulate the direction-changing manoeuvres following jump-landings, such as cutting sideways following a spike in volleyball, a header off in soccer, or a jump ball catch in American football.

Procedures

Participants first performed two standing and six practices to accustom the perturbation impact following a warm-up protocol (Song, Li, Layer, et al., 2023). For the standing practices, participants stood still with the ball contact at their lower-trunk locations on the left and right sides, respectively. For cutting practices, participants performed sidestep cutting tasks under upper-trunk and lower-trunk contralateral-perturbation, upper-trunk and lower-trunk ipsilateral-perturbation, as well as upper-trunk and lower-trunk no-perturbation conditions. Participants knew the perturbation directions and locations during practices.

Forty retro-reflective markers were placed on participants (the tip of the helmet, the seventh cervical vertebra, and bilateral gonions, acromioclavicular joints, olecranon processes, the

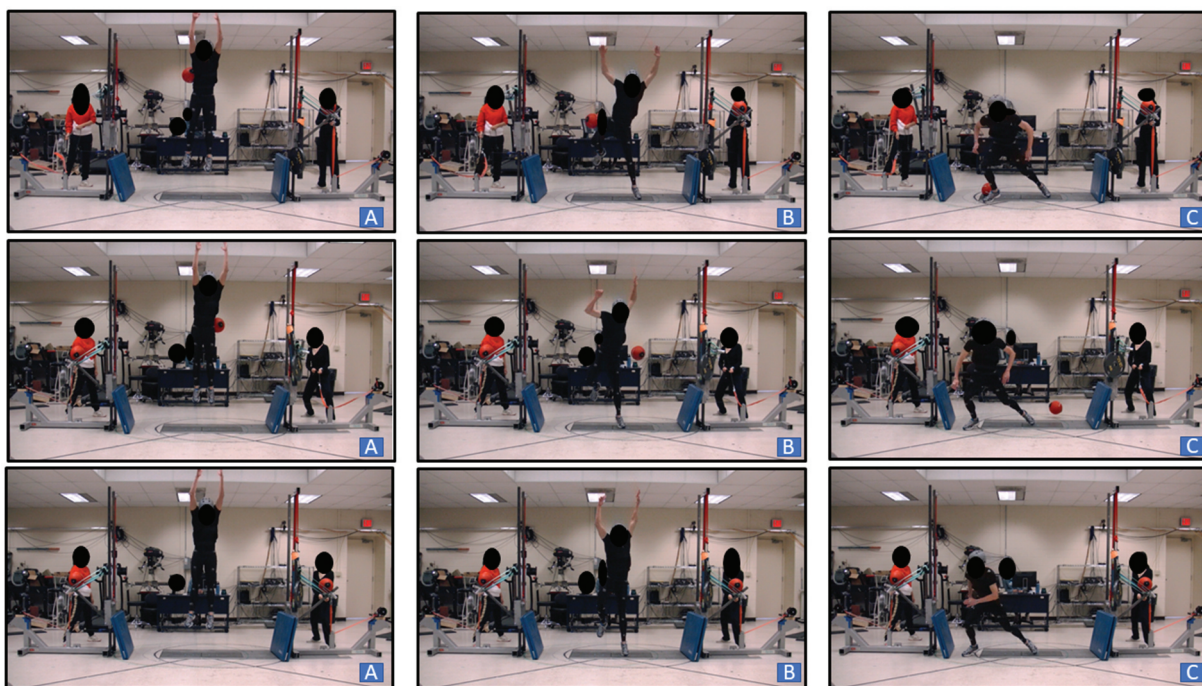


Figure 1. Upper trunk contralateral perturbation (top row), lower trunk ipsilateral perturbation (middle row), and lower trunk no perturbation (bottom row) when the participant landed with the left leg. Left column (A): ball contact near the maximal jump height; middle column (B): initial ground contact; right column (C): cutting after landing.

middle point of radial and ulnar styloid processes, third metacarpal heads, anterior superior iliac spine (ASIS), posterior superior iliac spine (PSIS), anterior thigh, lateral thigh, medial and lateral femoral condyles, tibia tuberosity, inferior shanks, medial and lateral malleolus, heels, tips of halluces, and first and fifth metatarsal heads), and two markers were placed on the diameter of each ball after the practices (Song, Li, Layer, et al., 2023). Eight opto-reflective cameras (Vicon Motion System, 160 Hz) were used to capture the three-dimensional coordinates of markers. A static trial was performed with participants standing still for 3 seconds, and then a minimum of three successful trials for each combination of perturbation locations and directions were performed in a randomised order. Participants knew the possible perturbation location but did not know the perturbation direction prior to official trials.

A successful trial was identified as 1) participants landed with the pre-determined foot on one force platform, 2) participants performed a 45-degree sidestep cutting with a maximum velocity for at least 5 metres, 3) the ball contacted the participants within 125 ms before/after the maximal jump height, and 4) the ball landed in a one-metre circle area after the contact with the participant to control the perturbation impact (Song, Li, Layer, et al., 2023). A trial was repeated if it did not meet all the requirements. Two subjective evaluations were assessed at the end of the testing, including whether participants could predict the perturbation direction during the official trials and how strong they felt the perturbation impact was compared to their experience of sports via a 5-point scale (Song, Li, Layer, et al., 2023, 2023).

Data reduction

Data reduction was performed in MATLAB 2022b. A fourth-order zero-lag Butterworth low-pass method was used to filter the coordinates of markers and GRF data at 15 Hz for the inverse dynamic approach (Kristianslund et al., 2012). GRF data were also separately filtered at 100 Hz for peak vertical GRF extraction. The relative positions of the four markers placed on the pelvis and each lower extremity segment were calibrated in the static trial and reconstructed by the four markers placed on the segment during the dynamic trials using the singular decomposition method (Soderkvist & Wedin, 1993). If markers were partially missing in a trial, the trajectory of a nearby marker placed on the same segment was used to fill the gap in the Vicon Nexus software. If a marker was entirely missing in a trial, the segment reference frame would be reconstructed with the other three markers placed on the segment. If more than one marker on the same segment was entirely missing in a trial, the trial was excluded from the analysis.

The whole-body COM was modelled by fifteen segments, including the head, upper trunk, pelvis, and bilateral upper arms, forearms, hands, thighs, shanks, and feet (Hinshaw et al., 2019). The hip joint centre was located in the pelvis reference frame, defined by the midpoint of the PSISs and bilateral ASISs, and 14% medially, 30% distally, and 22% posteriorly of the inter-ASIS distance to the ASIS for each side (Bell et al., 1989). The knee and ankle joint centres were defined as the midpoints of the medial and lateral femoral condyles and malleolus, respectively. An upper trunk segment reference frame was defined by the bilateral

acromioclavicular joints and the midpoint of the ASISs and PSISs. A thigh segment reference frame was determined using the hip joint centre, knee joint centre, and lateral femoral condyle. A shank segment reference frame was determined using the knee joint centre, ankle joint centre, and lateral malleolus. A foot segment reference frame was determined using the heel, tips of halluces, and fifth metatarsal head.

A bottom-up inverse dynamics approach was used to calculate three-dimensional internal knee joint resultant moments using anthropometric parameters described in a previous study (de Leva, 1996; Kingma et al., 1996). Cardan angles with an order of rotation of flexion (+)/extension (-), adduction (+)/abduction (-), and internal (+)/external (-) rotation were calculated between the thigh and shank segment reference frames for knee joint angles (Grood & Suntay, 1983). Cardan angles with the same order of rotation were calculated between the upper trunk segment and global reference frames for trunk angles. Joint angles during the static trials were subtracted from joint angles during dynamics trials. Knee moments were expressed in the shank reference frame as internal resultant moments (Song, Li, Layer, et al., 2023).

The dependent variables involved ball contact parameters (horizontal and vertical ball velocities at contact, timing offset between ball contact and maximum jump height, and contact location relative to the whole-body COM), performance variables (maximum jump height, stance time, and cutting velocity), and ACL loading variables. ACL loading variables included kinematic and kinetic variables at IC and during the landing phase (first 100 ms after IC) (Kristianslund et al., 2014; Song, Li, Layer, et al., 2023, 2023). Kinematic variables consisted of medial-lateral COM-Ankle distance, lateral trunk bending angle, knee flexion angle at IC, and peak knee flexion, abduction, and internal rotation angles during landing. Kinetic variables included peak VGRF, peak internal knee extension, adduction, and external rotation moments during landing. The VGRF was normalised by body weight, while the knee moments were normalised by the product of body weight and height.

Statistical analysis

Two- (upper-trunk vs. lower-trunk perturbation locations) by-two (contralateral-perturbation vs. ipsilateral-perturbation directions) repeated-measures analyses of variance (ANOVA) were conducted on ball velocities at contact, ball contact timing offset, and ball contact locations. Two- (upper-trunk vs. lower-trunk perturbation locations) by-three (no-perturbation, contralateral-perturbation, and ipsilateral-perturbation directions) repeated-measures ANOVAs were applied to ACL loading variables (medial-lateral COM-Ankle distance, lateral trunk bending angle, knee flexion angle at IC, peak knee flexion, abduction, and internal rotation angles, peak VGRF, and peak internal knee extension, adduction, and external rotation moments during landing) and performance variables (jump height, stance time, and cutting velocity). The sphericity of ANOVA was tested using Mauchly's Test. When the assumption of sphericity was violated (α less than 0.05), the Greenhouse-Geisser corrections were applied. Paired t-tests were performed between each pair of comparisons following

a significant main effect (Type I error rate of 0.05) detected by repeated-measures ANOVAs. The Benjamini-Hochberg procedure was performed on all t-tests to control the study-wide false discovery rate at 0.05 (Benjamini & Hochberg, 1995). Effect sizes were calculated by Cohen's d_z among all comparisons and considered as small (≤ 0.5), medium (0.5–0.8), and large (≥ 0.8) (Cohen, 1988). Consistent with previous studies (Song, Li, Layer, et al., 2023, 2023), secondary analyses were applied to ACL loading and performance variables, including sex as a between-subject factor, in analysing the effects of mid-flight external trunk perturbation locations and directions via two-by-two-by-three mixed-design ANOVAs. Statistical analyses were conducted in SPSS statistics 22.

Results

Nine trials (all from different conditions) were excluded from analyses due to the ball contacting participants too early/late (outside of the 125 ms window) or missing markers. The largest p-value was 0.003 following the false discovery adjustment. Males and females demonstrated similar responses to perturbation directions and locations in the secondary analyses (minimal interaction effects between sexes and perturbation conditions). The effect size and p-value of each comparison and the descriptive data with sex sub-groups are shown in the Appendixes.

Perturbation resulted in significantly slower cutting velocity, greater peak VGRF at both locations, and greater COM-Ankle distance only at the upper-trunk location (Tables 2 and 3). Ipsilateral perturbation resulted in the greatest peak knee internal rotation angles, least stance time, and greatest peak internal knee external rotation moments compared to other perturbation directions. For the upper-trunk location, contralateral perturbation resulted in the greatest lateral trunk bending angle (away from the travel direction), greatest peak knee flexion and abduction angles, and greatest peak internal knee adduction moment compared to other perturbation directions, while ipsilateral perturbation resulted in the trunk laterally bent to the travel direction, the smallest peak knee flexion angle, smallest peak knee abduction angle, and smallest peak internal knee adduction moment than other directions.

Upper-trunk perturbation showed greater jump height than the lower-trunk (Table 1). For ipsilateral perturbation, upper-trunk perturbation resulted in significantly greater COM-Ankle distance, trunk lateral bending towards the travel direction, smaller peak knee abduction angle, shorter stance time, greater peak internal knee extension moment, and smaller peak internal knee adduction moment compared to lower-trunk perturbation (Tables 2 and 3). For contralateral perturbation, upper-trunk perturbation demonstrated smaller COM-Ankle distance, greater trunk lateral bending away from the travel direction, greater peak knee flexion and abduction angles, smaller peak internal knee extension and external rotation moments, and greater peak internal knee adduction moment compared to lower-trunk perturbation.

Participants could not predict the perturbation directions during the testing. The subjective assessment of the perturbation was 2.8 ± 0.8 , suggesting a close-to-moderate perturbation (Song, Li, Layer, et al., 2023, 2023). The ball contacted

Table 1. Means $\pm$ standard deviations and p-values of repeated-measures ANOVAs for ball contact parameters.

				P-values for RMANOVAs		
		Ipsilateral perturbation	Contralateral perturbation	Location	Direction	Interaction
Horizontal Ball Velocity (m/s)	Upper Trunk	4.9 ± 0.2*	5.0 ± 0.3*	0.004	0.134	0.208
	Lower Trunk	5.0 ± 0.2*	5.1 ± 0.3*			
Vertical Ball Velocity (m/s)	Upper Trunk	0.02 ± 0.20 ^B	−0.22 ± 0.38 ^A	0.854	0.007	0.167
	Lower Trunk	−0.01 ± 0.19 ^B	−0.19 ± 0.32 ^A			
Timing Offset (ms)	Upper Trunk	−31.1 ± 32.3 ^A	−6.4 ± 41.0 ^B	0.039	0.002	0.224
	Lower Trunk	−17.1 ± 36.0 ^A	−1.3 ± 35.7 ^B			
Contact Location (m)	Upper Trunk	0.24 ± 0.07*	0.24 ± 0.06*	<0.001	0.989	0.992
	Lower Trunk	−0.01 ± 0.05*	−0.01 ± 0.07*			

Note. Timing offset: the time differences between the maximal jump height (calculated by whole-body centre of mass (COM)) and the ball contact, a negative number indicating the ball contacted earlier than the maximal jump height; Contact locations: the position relative to the whole-body COM, a positive number indicating the ball was higher than whole-body COM; *: significantly different between upper and lower trunk perturbation locations for each perturbation direction; ^A and ^B: ^A is significantly greater than ^B among perturbation directions for each perturbation location.

participants significantly higher with a smaller horizontal velocity in the upper-trunk conditions compared to the lower-trunk conditions, regardless of directions (Table 1). Ipsilateral perturbation contacted participants earlier than the maximum jump height with a smaller downward ball velocity compared to the contralateral perturbation at both locations.

Discussion

The purpose of this study was to investigate the effects of unanticipated mid-flight medial-lateral external trunk perturbation directions and locations on ACL loading variables and performance during sidestep cuttings. The findings supported the hypotheses that the contralateral perturbation to the upper-trunk would result in increased peak VGRF and peak internal knee adduction moments compared to the no perturbation condition, as well as increased peak knee abduction angles and stance time compared to the no perturbation and ipsilateral perturbation conditions but did not support increased sagittal-plane ACL loading variables during sidestep cuttings.

The contralateral perturbation applied a horizontal force and increased participants' horizontal velocity away from the cutting direction. As such, the participants placed the foot further away from the COM to stabilise and used a longer stance time to decelerate the increased horizontal velocity. For the upper-trunk conditions, as the force was applied superior to the COM, a torque was created to rotate the body away from the cutting direction. Consistently, the upper-trunk contralateral perturbation demonstrated the greatest trunk lateral bending away from the cutting direction. From a top-down perspective, a laterally bent trunk away from the cutting direction would require an increased internal hip adduction moment to stabilise the trunk, which may move the knee medially and increase internal knee adduction moments and knee abduction angles (Hinshaw et al., 2019). From a bottom-up perspective, a wider foot placement aligns with a more laterally located GRF vector, increasing the frontal-plane moment arm from the GRF to the knee (Fox, 2018). Although relatively large standard deviations were observed in peak knee abduction angles and internal knee adduction moments, the effect sizes of perturbation locations/directions on frontal-plane ACL loading variables were mostly greater than medium. Previous studies reported that female athletes showed greater

peak knee abduction angles and internal knee adduction moments during landing, which were associated with increased ACL injury risk (Hewett et al., 2005, 2009). In the current study, female and male participants demonstrated similar responses to the external perturbation. The more variable frontal-plane ACL loading variables might be explained by different cutting techniques among participants who have had different sports experiences. These findings were aligned with the literature showing that cutting with increased trunk lateral bending and foot placement away from the cutting direction was associated with increased frontal-plane ACL loading variables (greater knee abduction angles and internal knee adduction moments) (Dempsey et al., 2007; Donnelly et al., 2012; Kristianslund et al., 2014).

Yet, the greatest peak knee flexion angle was observed in the contralateral perturbation direction for both locations, which is associated with decreased sagittal-plane ACL loading (Song, Li, Layer, et al., 2023, 2023; Yom et al., 2014). While knee flexion angles at IC were similar between conditions, a greater peak knee flexion angle indicated a greater knee flexion range of motion (ROM) during landing (Song, Li, Layer, et al., 2023). Increased peak knee flexion angle allowed participants to absorb the impact forces over a greater ROM and decrease VGRF during landing (Dai et al., 2019). The increased knee flexion ROM appeared to be a compensatory strategy to prolong the landing phase to minimise the increases in loading variables. However, this soft-landing strategy also altered participants' preferred cutting strategies and resulted in longer stance time and slower cutting speed. Previous studies have documented potential trade-offs between ACL loading and performance variables (Dai et al., 2019; Dos'Santos et al., 2021), supporting participants to compensate for their performance to control the increases in knee loading in the current study. However, in real sports competitions, athletes might still need to maintain a short stance time and fast-cutting velocity after trunk perturbation. This will be a potentially risky scenario in which players may further experience increased ACL loading in sports. In contrast, ipsilateral perturbation to the upper trunk increased participants' horizontal velocity to the cutting direction and was the only condition that showed lateral trunk bending towards the cutting direction at IC. Therefore, the opposite changes are indicated by the decreased peak knee flexion angle, knee abduction angle, and internal knee adduction moment. Ipsilateral perturbation assisted in the

Table 2. Means $\pm$ standard deviations and p-values of repeated-measures ANOVAs for landing kinematic variables and performance variables.

		No perturbation	Ipsilateral perturbation	Contralateral perturbation	P-values for ANOVAs		
					Location	Direction	Interaction
Landing Kinematic Variables							
COM-Ankle Distance at Initial Contact (m)	Upper Trunk	0.16 ± 0.03 ^B	0.18 ± 0.04 ^{*A}	0.18 ± 0.03 ^{*A}	0.389	<0.001	<0.001
	Lower Trunk	0.16 ± 0.03 ^B	0.15 ± 0.03 ^{*B}	0.19 ± 0.04 ^{*A}			
Lateral Trunk Bending Angles at Initial Contact (°)	Upper Trunk	-3.4 ± 3.3 ^C	3.9 ± 3.9 ^{*B}	-8.2 ± 4.2 ^{*A}	0.049	<0.001	<0.001
	Lower Trunk	-3.3 ± 2.8 ^B	-5.8 ± 4.4 ^{*A}	-0.7 ± 3.6 ^{*C}			
Knee Flexion Angle at Initial Contact (°)	Upper Trunk	11.7 ± 6.3	12.3 ± 5.9	12.0 ± 6.0	0.636	0.167	0.802
	Lower Trunk	11.8 ± 6.8	12.6 ± 6.2	12.0 ± 6.0			
Peak Knee Flexion Angle during Landing (°)	Upper Trunk	48.7 ± 7.8 ^B	46.3 ± 7.6 ^C	52.0 ± 7.4 ^{*A}	0.939	<0.001	0.017
	Lower Trunk	48.7 ± 8.0 ^{AB}	48.1 ± 7.4 ^B	50.2 ± 7.1 ^{*A}			
Peak Knee Abduction Angle (-) during Landing (°)	Upper Trunk	-3.7 ± 3.5 ^B	-2.6 ± 3.2 ^{*C}	-4.5 ± 4.1 ^{*A}	0.778	<0.001	<0.001
	Lower Trunk	-3.7 ± 3.4	-3.7 ± 3.4 [*]	-3.7 ± 3.1 [*]			
Peak Knee Internal Rotation Angle during Landing (°)	Upper Trunk	4.0 ± 7.0 ^B	5.4 ± 7.5 ^A	3.9 ± 7.3 ^B	0.879	<0.001	0.660
	Lower Trunk	4.0 ± 6.5 ^B	6.0 ± 6.8 ^A	3.7 ± 6.8 ^B			
Performance Variables							
Jump Height (m)	Upper Trunk	0.44 ± 0.08 [*]	0.44 ± 0.09 [*]	0.44 ± 0.08 [*]	0.013	0.904	0.872
	Lower Trunk	0.43 ± 0.08 [*]	0.43 ± 0.08 [*]	0.43 ± 0.08 [*]			
Stance time (s)	Upper Trunk	0.49 ± 0.07 ^B	0.39 ± 0.07 ^{*C}	0.57 ± 0.11 ^A	0.792	<0.001	0.002
	Lower Trunk	0.49 ± 0.09 ^B	0.41 ± 0.07 ^{*C}	0.54 ± 0.14 ^A			
Cutting Velocity at Take-off (m/s)	Upper Trunk	2.3 ± 0.3 ^A	2.2 ± 0.4 ^B	2.2 ± 0.3 ^B	0.695	<0.001	0.458
	Lower Trunk	2.3 ± 0.4 ^A	2.2 ± 0.4 ^B	2.2 ± 0.4 ^B			

Note. Lateral trunk bending angles were positive when the trunk bent to the travel direction; COM-Ankle distance: absolute distance between the COM and the ankle centre of the landing leg in the medial-lateral direction; Stance time: the time between initial ground contact and take-off; Cutting velocity: the velocity at take-off after landing; ^A, ^B, and ^C: ^A was the greatest, ^B was the second greatest, ^C was the least among perturbation directions for each perturbation location (the conditions with the same letter were not significantly different from each other); *: significantly different between upper and lower perturbation locations for each perturbation direction.

Table 3. Means $\pm$ standard deviations and p-values of repeated-measures ANOVAs for landing kinetic variables.

		No perturbation	Ipsilateral perturbation	Contralateral perturbation	P-values for ANOVAs		
					Location	Direction	Interaction
Peak Vertical GRF during Landing (BW)	Upper Trunk	3.9 $\pm$ 0.8 ^B	4.2 $\pm$ 0.7 ^A	4.0 $\pm$ 0.8 ^A	0.609	<0.001	0.312
	Lower Trunk	3.8 $\pm$ 0.7 ^B	4.1 $\pm$ 0.7 ^A	4.1 $\pm$ 0.7 ^A			
Peak Knee Extension Moment (-) during Landing (BW*BH)	Upper Trunk	-0.13 $\pm$ 0.03 ^B	-0.14 $\pm$ 0.05 ^{*A}	-0.13 $\pm$ 0.03 ^{*AB}	0.994	0.083	0.001
	Lower Trunk	-0.13 $\pm$ 0.04 ^B	-0.13 $\pm$ 0.04 ^{*B}	-0.15 $\pm$ 0.04 ^{*A}			
Peak Knee Adduction Moment during Landing (BW*BH)	Upper Trunk	0.033 $\pm$ 0.023 ^B	0.027 $\pm$ 0.023 ^{*C}	0.045 $\pm$ 0.024 ^{*A}	0.390	0.149	<0.001
	Lower Trunk	0.030 $\pm$ 0.022 ^B	0.045 $\pm$ 0.026 ^{*A}	0.026 $\pm$ 0.019 ^{*B}			
Peak Knee External Rotation Moment (-) during Landing (BW*BH)	Upper Trunk	-0.018 $\pm$ 0.019 ^{*B}	-0.034 $\pm$ 0.033 ^A	-0.003 $\pm$ 0.004 ^{*C}	0.311	<0.001	0.048
	Lower Trunk	-0.014 $\pm$ 0.013 ^{*B}	-0.035 $\pm$ 0.044 ^A	-0.009 $\pm$ 0.008 ^{*C}			

Note. GRF: ground reaction force; BW: body weight; BH: body height; ^A, ^B, and ^C: ^A was the greatest, ^B was the second greatest, ^C was the least among perturbation directions for each perturbation location (the conditions with the same letter were not significantly different from each other); *: significantly different between upper and lower perturbation locations for each perturbation direction.

completion of the cutting motion and demonstrated the shortest stance time. In summary, the contralateral perturbation to the upper trunk resulted in increased frontal-plane ACL loading variables, and participants appeared to increase knee ROM and decrease cutting performance to compensate for control of these increases.

The perturbation also resulted in greater peak VGRF and slower cutting velocity at both locations, and the ipsilateral perturbation resulted in greater peak knee internal rotation angles and peak internal knee external rotation moments during landing at both locations. The increased peak VGRF is likely due to the perturbation applying an external force to individuals prior to landing, resulting in changes in whole-body COM horizontal velocity (Song, Li, Layer, et al., 2023, 2023; Yom et al., 2014). Therefore, participants landed with sub-optimal knee control and demonstrated slower cutting velocities. In addition, the upper-trunk perturbation location demonstrated greater jump height than the lower-trunk perturbation. This might be a compensatory strategy participants used to experience a lower possible perturbation position. As the upper trunk perturbation location was significantly superior to the whole-body COM and no significant differences were found in peak VGRF between locations, the slightly different jump heights were unlikely to confound the current findings.

Lastly, significant differences were found in ball contact parameters, but the differences in horizontal ball velocity were only 2% and likely caused by the inherent variation of the customised devices (Song, Li, Layer, et al., 2023). Additionally, smaller vertical ball velocity and earlier contact timing were found for the ipsilateral perturbation compared to the contralateral perturbation. Showing by lateral trunk bending at IC for the no perturbation conditions, participants typically bent their trunks away from the cutting direction in mid-flight as a countermovement before they bent their trunks towards the cutting direction during landing. As such, the trunk moved closer to the ipsilateral perturbation in mid-flight and experienced earlier contact in such a condition. On the other hand, the contralateral perturbation contacted the

participant slightly later when the ball was on its downward trajectory and, therefore, had a downward velocity. While these differences were statistically significant, the difference in vertical velocity was only 4% of the horizontal velocity, which still remained a major role in the medial-lateral perturbation (Song, Li, Layer, et al., 2023). Furthermore, the timing differences were 25 ms on average, less than 5% of the overall airborne phase. Overall, the perturbation was generally consistent among conditions, and the small differences were not likely to be significant confounding factors for the changes in ACL loading and performance variables, where changes with large effect sizes were observed.

The current study provides practical implications. The results provide direct connections between unanticipated trunk perturbation and ACL loading and performance variables during direction-changing manoeuvres. Although no direct force was applied to the knee, the perturbation changed the individual's movement pattern and self-selected strategies for sidestep cuttings. Compared to the effect of trunk perturbation during jump-landings, more studies are warranted in direction-changing manoeuvres to further quantify the potential trade-offs between ACL loading and performance variables when an external perturbation occurs. For example, ACL injury video analyses are suggested to quantify the cutting velocity, stance time, and surrounding environment when ACL tears during direction-changings. In addition, practitioners need to be aware of the unanticipated trunk perturbation during sidestep cuttings. Players might be encouraged to laterally bend their trunk to the travel direction or maintain a neutral trunk position, placing the foot close to the body and without extensive knee abduction during landing to prepare for such scenarios. These cutting techniques might be a strategy to optimise performance while minimising ACL loading (Dempsey et al., 2007; Donelon et al., 2020; Fox, 2018; Kristianslund et al., 2014), which might be incorporated into ACL injury screening tools and training programmes to improve ACL injury prevention. However, the training effects of cutting techniques on ACL

loading variables and performance, as well as injury prevention, need to be further investigated.

There were limitations in this study. First, although the differences in perturbation impact were unlikely to confound the results, a more consistent perturbation timing (at the take-off, near the maximum jump height, or at the IC) in different directions (sagittal plane) with a relative perturbation force (normalised by body weight) are worth studied during direction-changing manoeuvres. Second, participants performed approximately 6 repeated trials to achieve an overall 18 successful official trials in the current study. A larger force platform for landing with a greater testing area for a longer cutting distance after landing might allow participants to perform the task in a more real-life manner. Third, compared to the consistent findings of previous studies investigating trunk perturbation during jump-landings (Song, Li, Laver, et al., 2023; Yom et al., 2014), more studies are warranted to determine direction-changing manoeuvres with a more homogeneity sample to validate the findings of this study. Participants had different sports experiences (soccer, basketball, softball, etc.) in the current study, which might affect how they performed the cutting task. Meanwhile, only the 45-degree sidestep cutting was conducted with tape placed on the ground as guidance during the testing. Evaluating the effect of trunk perturbation on ACL loading variables and performance in various direction-changing tasks (crossover cutting and sidestep cutting in 30 degrees, 45 degrees, and 90 degrees) in a more natural cutting instruction (cutting direction guided through players and/or an object) aligned with different types of sports (volleyball, soccer, American football, etc.) is needed.

Conclusion

Mid-flight medial-lateral external pushing perturbation resulted in greater peak VGRF and slower cutting velocity during sidestep cuttings. Contralateral pushing perturbation to the upper trunk demonstrated the greatest peak knee abduction angles, peak internal knee adduction moments, and longer stance time. The increased frontal-plane ACL loading variables were likely caused by the increased lateral trunk bending and whole-body horizontal velocity away from the cutting direction. The findings may help understand the mechanisms of indirect contact ACL injuries in line with cutting performances and develop effective cutting techniques to prevent ACL injuries and improve performance during direction-changing manoeuvres.

Disclosure statement

No potential conflict of interest was reported by the author(s).

Funding

This work was supported by the National Science Foundation under grant [1933409]; International Society of Biomechanics in Sports under a student research grant; and National Institutes of Health under Wyoming INBRE [P20GM103432].

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Appendices

Appendix 1. Effect sizes (p-values) of paired comparison in ball contact parameters

	IP-UpT vs. IP-LoT	CP-UpT vs. CP-LoT	UpT-IP vs. UpT-CP	LoT-IP vs. LoT-CP
Horizontal Ball Velocity (m/s)	0.35 (0.004)	0.45 (0.004)	0.19 (0.134)	0.35 (0.134)
Vertical Ball Velocity (m/s)	0.27 (0.854)	0.15 (0.854)	0.54 (0.007)	0.45 (0.007)
Timing offset (ms)	0.40 (0.039)	0.18 (0.039)	0.54 (0.002)	0.49 (0.002)
Contact Location (m)	5.33 (<0.001)	5.08 (<0.001)	0.00 (0.989)	0.00 (0.989)

Note. Timing offset was defined as the time differences between the ball contact and the maximal centre of mass (COM) height with a negative number indicating the ball contacted early than the maximal COM height; Contact locations were defined as the differences between the ball and the COM in the vertical direction with a positive number indicating the ball was higher than COM; IP: ipsilateral perturbation; CP: contralateral perturbation; UpT: upper-trunk perturbation location; LoT: lower-trunk perturbation location. P-values of main effect has been recorded when there was no interaction.

Appendix 2. Effect sizes (p-values) of paired comparison in landing kinematic variables and performance variables

	NP-UpT vs. NP-LoT	IP-UpT vs. IP-LoT	CP-UpT vs. CP-LoT	UpT-NP vs. UpT-IP	UpT-NP vs. UpT-CP	UpT-IP vs. UpT-CP	LoT-NP vs. LoT-IP	LoT-NP vs. LoT-CP	LoT-IP vs. LoT-CP
Landing Kinematic Variables									
COM-Ankle Distance at Initial Contact (m)	0.04 (0.827)	0.79 (<0.001)	0.65 (<0.001)	0.62 (0.001)	0.74 (<0.001)	0.05 (0.776)	0.15 (0.395)	1.45 (<0.001)	1.21 (<0.001)
Lateral Trunk Bending Angles at Initial Contact (°)	0.02 (0.896)	2.11 (<0.001)	1.78 (<0.001)	1.64 (<0.001)	1.41 (<0.001)	2.88 (<0.001)	0.71 (<0.001)	0.94 (<0.001)	1.04 (<0.001)
Knee Flexion Angle at Initial Contact (°)	0.07 (0.636)	0.11 (0.636)	0.03 (0.636)	0.22 (0.167)	0.13 (0.167)	0.09 (0.167)	0.27 (0.167)	0.05 (0.167)	0.34 (0.167)
Peak Knee Flexion Angle during Early-Landing (°)	0.00 (0.984)	0.29 (0.110)	0.47 (0.012)	0.49 (0.010)	0.75 (<0.001)	1.00 (<0.001)	0.11 (0.530)	0.28 (0.119)	0.49 (0.010)
Peak Knee Abduction Angle (-) during Early-Landing (°)	0.00 (0.989)	0.64 (0.001)	0.54 (0.005)	0.71 (<0.001)	0.72 (<0.001)	0.97 (<0.001)	0.01 (0.955)	0.00 (0.992)	0.01 (0.964)
Peak Knee Internal Rotation Angle during Early-Landing (°)	0.00 (0.879)	0.12 (0.879)	0.06 (0.879)	0.43 (<0.001)	0.02 (0.656)	0.33 (0.003)	0.59 (<0.001)	0.08 (0.656)	0.55 (0.003)
Performance Variables									
Jump Height (m)	0.35 (0.013)	0.33 (0.013)	0.38 (0.013)	0.09 (0.904)	0.02 (0.904)	0.08 (0.904)	0.01 (0.904)	0.06 (0.904)	0.05 (0.904)
Stance time (s)	0.10 (0.589)	0.48 (0.011)	0.29 (0.112)	1.56 (<0.001)	0.86 (<0.001)	2.20 (<0.001)	1.07 (<0.001)	0.68 (<0.001)	1.34 (<0.001)
Cutting Velocity at Take-off (m/s)	0.15 (0.695)	0.10 (0.695)	0.14 (0.695)	0.58 (<0.001)	0.86 (<0.001)	0.11 (0.929)	0.42 (<0.001)	0.85 (<0.001)	0.10 (0.929)

Note. Lateral trunk bending angles were positive when the trunk bent to the travel direction; COM-ankle distance was defined as the absolute distance between the whole-body centre of mass (COM) and the ankle centre of the landing leg in the medial-lateral direction; NP: no perturbation; IP: ipsilateral perturbation; CP: contralateral perturbation; UpT: upper-trunk perturbation location; LoT: lower-trunk perturbation location. P-values of main effect has been recorded when there was no interaction.

Appendix 3. Effect sizes (p-values) of paired comparison in landing kinetic variables

	NP-UpT vs. NP-LoT	IP-UpT vs. IP-LoT	CP-UpT vs. CP-LoT	UpT-NP vs. UpT-IP	UpT-NP vs. UpT-CP	UpT-IP vs. UpT-CP	LoT-NP vs. LoT-IP	LoT-NP vs. LoT-CP	LoT-IP vs. LoT-CP
Peak Vertical GRF during Early-Landing (BW)	0.12 (0.609)	0.18 (0.609)	0.20 (0.609)	0.66 (<0.001)	0.20 (0.028)	0.41 (0.077)	0.55 (<0.001)	0.58 (0.028)	0.11 (0.077)
Peak Knee Extension Moment (-) during Early-Landing (BW*BH)	0.05 (0.765)	0.41 (0.026)	0.52 (0.006)	0.48 (0.011)	0.10 (0.578)	0.30 (0.098)	0.06 (0.717)	0.57 (0.003)	0.59 (0.002)
Peak Knee Adduction Moment during Early-Landing (BW*BH)	0.29 (0.107)	1.19 (<0.001)	1.02 (<0.001)	0.40 (0.030)	0.92 (<0.001)	0.92 (<0.001)	0.87 (<0.001)	0.24 (0.188)	0.75 (<0.001)
Peak Knee External Rotation Moment (-) during Early-Landing (BW*BH)	0.42 (0.024)	0.06 (0.722)	0.79 (<0.001)	0.95 (<0.001)	0.79 (<0.001)	0.94 (<0.001)	0.63 (0.001)	0.45 (0.016)	0.63 (0.001)

Note. GRF: ground reaction force; BW: body weight; BH: body height; NP: no perturbation; IP: ipsilateral perturbation; CP: contralateral perturbation; UpT: upper-trunk perturbation location; LoT: lower-trunk perturbation location. P-values of main effect has been recorded when there was no interaction.

Appendix 4. Means $\pm$ standard deviations for landing kinematic variables and performance variables with sex sub-groups

			No perturbation	Ipsilateral perturbation	Contralateral perturbation
Landing Kinematic Variables					
COM-Ankle Distance at Initial Contact (m)	Upper Trunk	Female	0.16 ± 0.03	0.17 ± 0.05	0.18 ± 0.04
		Male	0.16 ± 0.03	0.18 ± 0.03	0.18 ± 0.03
	Lower Trunk	Female	0.15 ± 0.04	0.14 ± 0.04	0.19 ± 0.04
		Male	0.16 ± 0.03	0.16 ± 0.02	0.20 ± 0.03
Lateral Trunk Bending Angles at Initial Contact (°)	Upper Trunk	Female	−2.1 ± 2.7	4.7 ± 4.7	−7.0 ± 3.6
		Male	−4.6 ± 3.4	3.2 ± 2.9	−9.3 ± 4.4
	Lower Trunk	Female	−2.7 ± 2.6	−5.7 ± 5.3	0.2 ± 4.4
		Male	−3.9 ± 2.9	−5.8 ± 3.6	−1.6 ± 2.5
Knee Flexion Angle at Initial Contact (°)	Upper Trunk	Female	10.3 ± 5.6	12.1 ± 5.1	10.7 ± 4.6
		Male	12.9 ± 6.9	12.5 ± 6.7	13.2 ± 7.0
	Lower Trunk	Female	11.0 ± 6.4	11.6 ± 4.9	11.0 ± 4.8
		Male	12.5 ± 7.2	13.5 ± 7.1	12.8 ± 6.9
Peak Knee Flexion Angle during Early-Landing (°)	Upper Trunk	Female	46.4 ± 6.5	43.2 ± 7.5	49.9 ± 7.4
		Male	50.7 ± 8.4	49.1 ± 6.9	53.9 ± 7.1
	Lower Trunk	Female	46.2 ± 6.0	44.3 ± 6.5	48.6 ± 6.3
		Male	50.9 ± 9.0	51.4 ± 6.6	51.5 ± 7.6
Peak Knee Abduction Angle (-) during Early-Landing (°)	Upper Trunk	Female	−4.2 ± 3.0	−2.6 ± 2.6	−5.1 ± 4.3
		Male	−3.1 ± 3.8	−2.7 ± 3.6	−4.1 ± 3.9
	Lower Trunk	Female	−3.7 ± 3.1	−3.8 ± 3.0	−3.8 ± 3.0
		Male	−3.6 ± 3.7	−3.6 ± 3.8	−3.6 ± 3.3
Peak Knee Internal Rotation Angle during Early-Landing (°)	Upper Trunk	Female	5.8 ± 6.8	6.1 ± 6.6	5.8 ± 6.6
		Male	2.4 ± 6.9	4.8 ± 8.3	2.3 ± 7.7
	Lower Trunk	Female	5.0 ± 6.1	6.8 ± 6.3	5.1 ± 7.2
		Male	3.1 ± 6.9	5.2 ± 7.2	2.4 ± 6.5
Performance Variables					
Jump Height (m)	Upper Trunk	Female	0.37 ± 0.05	0.37 ± 0.05	0.37 ± 0.04
		Male	0.50 ± 0.06	0.50 ± 0.07	0.49 ± 0.07
	Lower Trunk	Female	0.37 ± 0.05	0.37 ± 0.05	0.36 ± 0.05
		Male	0.49 ± 0.06	0.49 ± 0.07	0.49 ± 0.06
Stance time (s)	Upper Trunk	Female	0.49 ± 0.10	0.39 ± 0.09	0.58 ± 0.14
		Male	0.50 ± 0.05	0.39 ± 0.04	0.55 ± 0.08
	Lower Trunk	Female	0.49 ± 0.12	0.43 ± 0.09	0.57 ± 0.19
		Male	0.48 ± 0.06	0.40 ± 0.05	0.52 ± 0.06
Cutting Velocity at Take-off (m/s)	Upper Trunk	Female	2.2 ± 0.2	2.0 ± 0.3	2.1 ± 0.2
		Male	2.5 ± 0.3	2.4 ± 0.3	2.4 ± 0.4
	Lower Trunk	Female	2.1 ± 0.3	2.1 ± 0.3	2.0 ± 0.3
		Male	2.5 ± 0.3	2.3 ± 0.4	2.4 ± 0.3

Note. COM: centre of mass.

Appendix 5. Means $\pm$ standard deviations for landing kinetic variables with sex sub-groups

			No perturbation	Ipsilateral perturbation	Contralateral perturbation
Peak Vertical GRF during Early-Landing (BW)	Upper Trunk	Female	3.6 $\pm$ 0.7	3.9 $\pm$ 0.5	3.9 $\pm$ 0.8
		Male	4.2 $\pm$ 0.9	4.5 $\pm$ 0.8	4.1 $\pm$ 0.7
	Lower Trunk	Female	3.6 $\pm$ 0.6	3.9 $\pm$ 0.6	3.9 $\pm$ 0.7
		Male	4.1 $\pm$ 0.8	4.3 $\pm$ 0.8	4.3 $\pm$ 0.7
Peak Knee Extension Moment (-) during Early-Landing (BW*BH)	Upper Trunk	Female	-0.12 $\pm$ 0.03	-0.13 $\pm$ 0.05	-0.13 $\pm$ 0.03
		Male	-0.14 $\pm$ 0.03	-0.16 $\pm$ 0.05	-0.14 $\pm$ 0.03
	Lower Trunk	Female	-0.13 $\pm$ 0.04	-0.12 $\pm$ 0.04	-0.14 $\pm$ 0.03
		Male	-0.14 $\pm$ 0.04	-0.14 $\pm$ 0.03	-0.15 $\pm$ 0.04
Peak Knee Adduction Moment during Early-Landing (BW*BH)	Upper Trunk	Female	0.029 $\pm$ 0.019	0.022 $\pm$ 0.018	0.039 $\pm$ 0.018
		Male	0.036 $\pm$ 0.025	0.031 $\pm$ 0.027	0.051 $\pm$ 0.028
	Lower Trunk	Female	0.024 $\pm$ 0.017	0.042 $\pm$ 0.023	0.022 $\pm$ 0.017
		Male	0.035 $\pm$ 0.025	0.048 $\pm$ 0.028	0.031 $\pm$ 0.019
Peak Knee External Rotation Moment (-) during Early-Landing (BW*BH)	Upper Trunk	Female	-0.019 $\pm$ 0.022	-0.034 $\pm$ 0.034	-0.002 $\pm$ 0.002
		Male	-0.017 $\pm$ 0.016	-0.035 $\pm$ 0.034	-0.004 $\pm$ 0.005
	Lower Trunk	Female	-0.013 $\pm$ 0.012	-0.037 $\pm$ 0.050	-0.009 $\pm$ 0.008
		Male	-0.015 $\pm$ 0.014	-0.034 $\pm$ 0.039	-0.008 $\pm$ 0.008

Note. GRF: ground reaction force; BW: body weight; BH: body height.